

## **From Industry 4.0 to the Industry 5.0 paradigm**

### **Initial Post**

Technological advances have become an integral part of our daily lives and are rapidly evolving across diverse domains. Machine-to-machine interactions disrupt traditional workflows and processes, driving cost-efficient automation and productivity at scale, which are key gains of Industry 4.0. While this era provides the digital foundations—from robotics and the Internet of Things to advancements in artificial intelligence (Schwab, 2016)—it does not offer a comprehensive approach to managing these hyperconnected systems or addressing their failures. This is where Industry 5.0 steps in as a response, focusing not only on correcting and controlling the risks created by 'smart' systems, but also placing societal wellbeing, system resilience, and sustainability as a central part of the technological progress (European Commission, 2021).

A notable example is the cyberattack on MGM Resorts and Casinos in Las Vegas, which demonstrates how the rapid advances of Industry 4.0—system digitalisation, seamless connectivity, and efficiency-driven automation—can also introduce critical vulnerabilities. The attack began when a single weak human element was exploited through a phishing call, granting hackers access to MGM's entire information system. This breach paralysed the company's high-tech infrastructure, halted operations across the enterprise, and forced a return to manual, paper-and-pen administration. Most significantly, the incident exposed sensitive customer data, undermining trust and privacy for millions. In addition to this customer impact, MGM suffered \$100 million in lost revenue, severe reputational damage, public scrutiny, and legal consequences (Collier, 2023; Weisman, 2025).

MGM did not account for fallbacks or the resilience-focused design advocated by Industry 5.0. Human involvement in workflows alone does not guarantee robust human-to-machine interaction. As Metcalf (2024) notes, prioritising efficiency leaves little space for responsiveness and adaptability when facing adversarial threats. Post hoc corrections are often insufficient for effective recovery and minimising disruption.

The MGM case highlights that while efficiency and productivity are necessary, they are not sufficient. Moreover, artificial intelligence and advanced algorithms do not solve these challenges on their own, but they can amplify gaps in existing infrastructure. Only intelligent human oversight, supported by clear accountability frameworks and resilience measures, can help contain the negative impacts of technological transformation. These elements must be integral to hyperconnected system designs to safeguard human wellbeing, promote human values, and build societal trust for long-term economic progress and prosperity.

## References

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## Summary Post

My initial post highlighted the gap between Industry 4.0 achievements and the requirements of the Industry 5.0 paradigm, stressing that pursuing cost-cutting efficiency gains is no longer sufficient. The ransomware attacks on MGM Hospitality underscored that while technological capabilities unlock systems automation and hyperconnectivity and bring undoubted economic benefits, they cannot stand alone. As Metcalf (2024) states, these must integrate resilience and human-centric values to respond to socio-technical challenges.

The peer feedback has emphasised that prevention differs from resilience in modern risk management. Standard security measures such as verification, least privilege, and network segmentation offer preventive protection and enable quicker recovery. However, as Alkove (2021) argues, companies should treat adversary attacks as business as usual and focus on when, not if, they will occur. Enterprises should design adaptive systems, responsive processes, accountability guidance, and resilient architecture that keep pace with technological threats to sustainably protect business continuity and privacy.

The discussion also deepened my understanding of the impact on the underlying data and operations. Manual logs and offline backups can provide safeguards, but they also raise important questions about data integrity. Reconciliation and data quality validation are essential to ensure that reporting blackout periods, potential data deletion, insertion and corruption across business units are captured. This process helps ensure the data is trustworthy before resuming operations, restarting model-based personalised services, and building further predictions. Exploratory data analysis can help detect anomalies and inconsistencies in the post-recovery stage, but as Peng, Hahn and Huang (2022) observe, incomplete or improperly restored data can alter variable distribution and relationships, increasing potential

bias in variance and correlation. As Campbell (2026) points out, even minor inaccuracies can be intensified by Artificial Intelligence, causing substantial business harm.

Overall, Industry 5.0 promotes an advanced ecosystem spanning technology, data integrity, and operational reliability that should be accompanied by human critical oversight, sound judgment, and ongoing feedback loops. Enforcing these measures enables well-informed decision-making, protects human well-being, and promotes societal trust in the underlying system.

## References

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